Z	Symbol	Element	IP_1 (eV)	$Y_{\mathrm{IP}} = 1/\mathrm{IP}_1$	$(q_2, m_2)_{\mathrm{IP}}$
1	H	Hydrogen	13.59844	0.073539	$(3,1)_{\mathrm{IP}}$
2	He	Helium	24.58738	0.040672	$(4,3)_{\mathrm{IP}}$
3	Li	Lithium	5.39171	0.185466	$(3,3)_{\mathrm{IP}}$
4	Be	Beryllium	9.32269	0.107267	$(3,2)_{\mathrm{IP}}$
5	B	Boron	8.29803	0.120511	$(3,2)_{\mathrm{IP}}$
6	C	Carbon	11.26030	0.088806	$(3,2)_{\mathrm{IP}}$
7	N	Nitrogen	14.53414	0.068804	$(3,1)_{\mathrm{IP}}$
8	O	Oxygen	13.61806	0.073429	$(3,1)_{\mathrm{IP}}$
9	F	Fluorine	17.42282	0.057395	$(3,1)_{\mathrm{IP}}$
10	Ne	Neon	21.5646	0.046373	$(4,3)_{\mathrm{IP}}$
11	Na	Sodium	5.13908	0.194587	$(3,4)_{\mathrm{IP}}$
12	Mg	Magnesium	7.64624	0.130782	$(3,2)_{\mathrm{IP}}$
13	Al	Aluminium	5.98577	0.167043	$(3,3)_{\mathrm{IP}}$
14	Si	Silicon	8.15169	0.122680	$(3,2)_{\mathrm{IP}}$
15	P	Phosphorus	10.48669	0.095359	$(3,2)_{\mathrm{IP}}$
16	S	Sulfur	10.36001	0.096525	$(3,2)_{\mathrm{IP}}$
17	Cl	Chlorine	12.96764	0.077115	$(3,1)_{\mathrm{IP}}$
18	Ar	Argon	15.75962	0.063453	$(3,1)_{\mathrm{IP}}$
19	K	Potassium	4.34066	0.230382	$(3,4)_{\mathrm{IP}}$
20	Ca	Calcium	6.11316	0.163578	$(3,3)_{\mathrm{IP}}$
21	Sc	Scandium	6.5615	0.152404	$(3,3)_{\mathrm{IP}}$
22	Ti	Titanium	6.8281	0.146448	$(3,3)_{\mathrm{IP}}$
23	V	Vanadium	6.7462	0.148227	$(3,3)_{\mathrm{IP}}$
24	Cr	Chromium	6.7665	0.147781	$(3,3)_{\mathrm{IP}}$
25	Mn	Manganese	7.43402	0.134518	$(3,2)_{\mathrm{IP}}$
26	Fe	Iron	7.9024	0.126544	$(3,2)_{\mathrm{IP}}$
27	Co	Cobalt	7.8810	0.126888	$(3,2)_{\mathrm{IP}}$
28	Ni	Nickel	7.6398	0.130893	$(3,2)_{\mathrm{IP}}$
29	Cu	Copper	7.72638	0.129426	$(3,2)_{\mathrm{IP}}$
30	Zn	Zinc	9.3942	0.106447	$(3,2)_{\mathrm{IP}}$

TABLE 1. First ionization potentials (NIST ASD 2023), transformed values $Y_{\mathrm{IP}} = 1/\mathrm{IP}$, and dominant descriptors $(q_2, m_2)_{\mathrm{IP}}$ determined by the HAA algorithm for elements Z=1–30.

Z	Symbol	Element	IP_1 (eV)	$Y_{\mathrm{IP}} = 1/\mathrm{IP}_1$	$(q_2, m_2)_{\mathrm{IP}}$
31	Ga	Gallium	5.99930	0.166680	$(3,3)_{\mathrm{IP}}$
32	Ge	Germanium	7.8994	0.126590	$(3,2)_{\mathrm{IP}}$
33	As	Arsenic	9.7886	0.102160	$(3,2)_{\mathrm{IP}}$
34	Se	Selenium	9.75238	0.102540	$(3,2)_{\mathrm{IP}}$
35	Br	Bromine	11.81381	0.084645	$(3,2)_{\mathrm{IP}}$
36	Kr	Krypton	13.99961	0.071429	$(3,1)_{\mathrm{IP}}$
37	Rb	Rubidium	4.17713	0.239399	$(3,4)_{\mathrm{IP}}$
38	Sr	Strontium	5.6949	0.175594	$(3,3)_{\mathrm{IP}}$
39	Y	Yttrium	6.2171	0.160847	$(3,3)_{\mathrm{IP}}$
40	Zr	Zirconium	6.63390	0.150742	$(3,3)_{\mathrm{IP}}$
41	Nb	Niobium	6.75885	0.147946	$(3,3)_{\mathrm{IP}}$
42	Mo	Molybdenum	7.09243	0.140994	$(3,3)_{\mathrm{IP}}$
43	Tc	Technetium	7.28	0.137363	$(3,3)_{\mathrm{IP}}$
44	Ru	Ruthenium	7.36050	0.135863	$(3,3)_{\mathrm{IP}}$
45	Rh	Rhodium	7.45890	0.134068	$(3,3)_{\mathrm{IP}}$
46	Pd	Palladium	8.3369	0.119948	$(3,2)_{\mathrm{IP}}$
47	Ag	Silver	7.5762	0.131991	$(3,2)_{\mathrm{IP}}$
48	Cd	Cadmium	8.9938	0.111187	$(3,2)_{\mathrm{IP}}$
49	In	Indium	5.78636	0.172816	$(3,3)_{\mathrm{IP}}$
50	Sn	Tin	7.3439	0.136165	$(3,2)_{\mathrm{IP}}$
51	Sb	Antimony	8.6084	0.116166	$(3,2)_{\mathrm{IP}}$
52	Te	Tellurium	9.0096	0.110994	$(3,2)_{\mathrm{IP}}$
53	I	Iodine	10.45126	0.095682	$(3,2)_{\mathrm{IP}}$
54	Xe	Xenon	12.1298	0.082442	$(3,2)_{\mathrm{IP}}$
55	Cs	Caesium	3.89390	0.256814	$(3,4)_{\mathrm{IP}}$
56	Ba	Barium	5.21170	0.191876	$(3,4)_{\mathrm{IP}}$
57	La	Lanthanum	5.5769	0.179306	$(3,3)_{\mathrm{IP}}$
58	Ce	Cerium	5.5387	0.180547	$(3,3)_{\mathrm{IP}}$
59	Pr	Praseodymium	5.473	0.182715	$(3,3)_{\mathrm{IP}}$
60	Nd	Neodymium	5.5250	0.180995	$(3,3)_{\mathrm{IP}}$

TABLE 2. First ionization potentials (NIST ASD 2023), transformed values $Y_{\mathrm{IP}} = 1/\mathrm{IP}$, and dominant descriptors $(q_2, m_2)_{\mathrm{IP}}$ determined by the HAA algorithm for elements Z=31–60.

Z	Symbol	Element	IP_1 (eV)	$Y_{\mathrm{IP}} = 1/\mathrm{IP}_1$	$(q_2, m_2)_{\mathrm{IP}}$
61	Pm	Promethium	5.582	0.179147	$(3,3)_{\mathrm{IP}}$
62	Sm	Samarium	5.6436	0.177189	$(3,3)_{\mathrm{IP}}$
63	Eu	Europium	5.6704	0.176354	$(3,3)_{\mathrm{IP}}$
64	Gd	Gadolinium	6.1501	0.162599	$(3,3)_{\mathrm{IP}}$
65	Tb	Terbium	5.8638	0.170540	$(3,3)_{\mathrm{IP}}$
66	Dy	Dysprosium	5.9389	0.168380	$(3,3)_{\mathrm{IP}}$
67	Ho	Holmium	6.0215	0.166067	$(3,3)_{\mathrm{IP}}$
68	Er	Erbium	6.1077	0.163727	$(3,3)_{\mathrm{IP}}$
69	Tm	Thulium	6.18431	0.161699	$(3,3)_{\mathrm{IP}}$
70	Yb	Ytterbium	6.25416	0.159893	$(3,3)_{\mathrm{IP}}$
71	Lu	Lutetium	5.4259	0.184300	$(3,3)_{\mathrm{IP}}$
72	Hf	Hafnium	6.82507	0.146521	$(3,3)_{\mathrm{IP}}$
73	Ta	Tantalum	7.5496	0.132456	$(3,3)_{\mathrm{IP}}$
74	W	Tungsten	7.8640	0.127163	$(3,2)_{\mathrm{IP}}$
75	Re	Rhenium	7.8335	0.127656	$(3,2)_{\mathrm{IP}}$
76	Os	Osmium	8.4382	0.118508	$(3,2)_{\mathrm{IP}}$
77	Ir	Iridium	8.9670	0.111520	$(3,2)_{\mathrm{IP}}$
78	Pt	Platinum	8.9587	0.111625	$(3,2)_{\mathrm{IP}}$
79	Au	Gold	9.2255	0.108396	$(3,2)_{\mathrm{IP}}$
80	Hg	Mercury	10.43750	0.095808	$(3,2)_{\mathrm{IP}}$
81	Tl	Thallium	6.1082	0.163714	$(3,3)_{\mathrm{IP}}$
82	Pb	Lead	7.41666	0.134831	$(3,2)_{\mathrm{IP}}$
83	Bi	Bismuth	7.2856	0.137257	$(3,2)_{\mathrm{IP}}$
84	Po	Polonium	8.417	0.118807	$(3,2)_{\mathrm{IP}}$
85	At	Astatine	*920 (WEL)*	∼0.1087	$(3,2)_{\mathrm{IP}}$
86	Rn	Radon	10.74850	0.093036	$(3,2)_{\mathrm{IP}}$
87	Fr	Francium	4.0727	0.245538	$(3,4)_{\mathrm{IP}}$
88	Ra	Radium	5.2784	0.189449	$(3,4)_{\mathrm{IP}}$
89	Ac	Actinium	5.17	0.193424	$(3,4)_{\mathrm{IP}}$
90	Th	Thorium	6.3067	0.158562	$(3,3)_{\mathrm{IP}}$

TABLE 3. First ionization potentials (NIST ASD 2023), transformed values $Y_{\mathrm{IP}} = 1/\mathrm{IP}$, and dominant descriptors $(q_2, m_2)_{\mathrm{IP}}$ determined by the HAA algorithm for elements Z=61–90. The value for astatine (At) is a theoretical estimate (WEL).

Z	Symbol	Element	IP_1 (eV)	$Y_{\mathrm{IP}} = 1/\mathrm{IP}_1$	$(q_2, m_2)_{\mathrm{IP}}$
91	Pa	Protactinium	5.89	0.169779	$(3,3)_{\mathrm{IP}}$
92	U	Uranium	6.19405	0.161445	$(3,3)_{\mathrm{IP}}$
93	Np	Neptunium	6.2657	0.159599	$(3,3)_{\mathrm{IP}}$
94	Pu	Plutonium	6.0262	0.165941	$(3,3)_{\mathrm{IP}}$
95	Am	Americium	5.9738	0.167395	$(3,3)_{\mathrm{IP}}$
96	Cm	Curium	5.9915	0.166883	$(3,3)_{\mathrm{IP}}$
97	Bk	Berkelium	6.1979	0.161345	$(3,3)_{\mathrm{IP}}$
98	Cf	Californium	6.2817	0.159192	$(3,3)_{\mathrm{IP}}$
99	Es	Einsteinium	6.42	0.155763	$(3,3)_{\mathrm{IP}}$
100	Fm	Fermium	6.50	0.153846	$(3,3)_{\mathrm{IP}}$
101	Md	Mendelevium	6.58	0.151976	$(3,3)_{\mathrm{IP}}$
102	No	Nobelium	6.65	0.150376	$(3,3)_{\mathrm{IP}}$
103	Lr	Lawrencium	4.9	0.204082	$(3,4)_{\mathrm{IP}}$
104	Rf	Rutherfordium	6.0	0.166667	$(3,3)_{\mathrm{IP}}$

TABLE 4. First ionization potentials (NIST ASD 2023), transformed values $Y_{\mathrm{IP}} = 1/\mathrm{IP}$, and dominant descriptors $(q_2, m_2)_{\mathrm{IP}}$ determined by the HAA algorithm for transuranium elements Z=91–104.

Dominant level q_2	$C \in [\,,\,)$
$q_2 = 2$	$[2.73240 \times 10^{-1}, 1)$
$q_2 = 3$	$[5.39457 \times 10^{-2}, 2.73240 \times 10^{-1})$
$q_2 = 4$	$[1.48657 \times 10^{-2}, 5.39457 \times 10^{-2})$
$q_2 = 5$	$[4.54259 \times 10^{-3}, 1.48657 \times 10^{-2})$
$q_2 = 6$	$[1.44907 \times 10^{-3}, 4.54259 \times 10^{-3})$
$q_2 = 7$	$[4.71765 \times 10^{-4}, 1.44907 \times 10^{-3})$
$q_2 = 8$	$[1.55203 \times 10^{-4}, 4.71765 \times 10^{-4})$
$q_2 = 9$	$[5.13478 \times 10^{-5}, 1.55203 \times 10^{-4})$
$q_2 = 10$	$[1.70417 \times 10^{-5}, 5.13478 \times 10^{-5})$

TABLE 5. Classification map for the hierarchical level q_2 for real numbers $C \in (0, 1)$.